

Lesson 04 · Collections, LINQ & Functional C#

Java Streams, Kotlin collections and TypeScript array methods, re-mapped onto .NET collections, LINQ, delegates and expression trees

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Page numbers are resolved from the rendered PDF. Code panels are cut from the compiled samples.

1 · Why collections and LINQ decide how .NET code reads

Most business C# is collection code: load some records, filter, group, aggregate, map to a response — and LINQ is the vocabulary every reviewer expects you to read at a glance. You already think in pipelines: Java Streams and Collectors in Spring services, Kotlin collection operators, TypeScript array methods in NestJS and React, and a few hundred dashboard queries for a BI tool. Most operators map one-to-one. This lesson spends its words on the ones that do not.

Three things do not transfer, and they are where production bugs come from. A LINQ query over `IEnumerable<T>` is re-runnable and lazy, so it silently runs again every time it is enumerated. A lambda can compile to executable code *or* to a data structure a provider translates to SQL or a DynamoDB expression, depending only on the parameter type. And collection interfaces make promises — read-only, immutable, frozen — that are easy to confuse.

You will build the tool you already built once in another language: a query generator. The samples turn a C# predicate into a DynamoDB filter expression with `#name` and `:value` placeholders, run MotorQuote dashboard queries over a deterministic quote book, and repeat the dashboard in Visual Basic query syntax — the form you will meet in older estates.

1.1 · The numbers to leave this lesson with

ONE QUERY, THREE
PASSES

482 calls

predicate calls over 240
quotes · measured

NEW IN .NET 9 LINQ

3

CountBy · AggregateBy ·
Index · verified

VB-ONLY QUERY
CLAUSES

6

no C# query keyword ·
verified

DYNAMODB IN LIST

≤ 100

values per IN operator ·
verified

LESSON 04 SAMPLES

873 lines

7 projects · 41 tests ·
measured

KEY: • Same idea • Renamed • Different • Trap • Verified • Estimate • Measured C# Java/Kotlin architecture

i Before you start

- You need lesson 02 (C# Language Essentials) and lesson 03 (Types, OOP & Generics).** Records, with, generics and extension methods are used here without explanation; LINQ operators *are* extension methods on `IEnumerable<T>`.
- You will build seven projects.** L04.QuoteData (MotorQuote records, a deterministic 240-quote book, dashboard queries, rating rules as functions), L04.QueryGen (expression tree → DynamoDB filter), two C# consoles (L04.Collections, L04.QuoteAnalytics), a VB console (L04.VbLinq) and two xUnit projects. Premiums are illustrative, not a real tariff.
- Not here:** `IAsyncEnumerable<T>` and PLINQ are lesson 05 (Async, Tasks & Concurrency); how EF Core translates `IQueryable` to SQL is lesson 09 (Data Access with EF Core); VB language semantics in depth are lesson 06 (VB.NET for Legacy Estates).
- Facts that change with releases are marked • Verified and link to the source; figures computed from the samples are • Measured; Java, Kotlin and TypeScript panels are for comparison and are not compiled.

2 · The collection family — what to accept, store and return

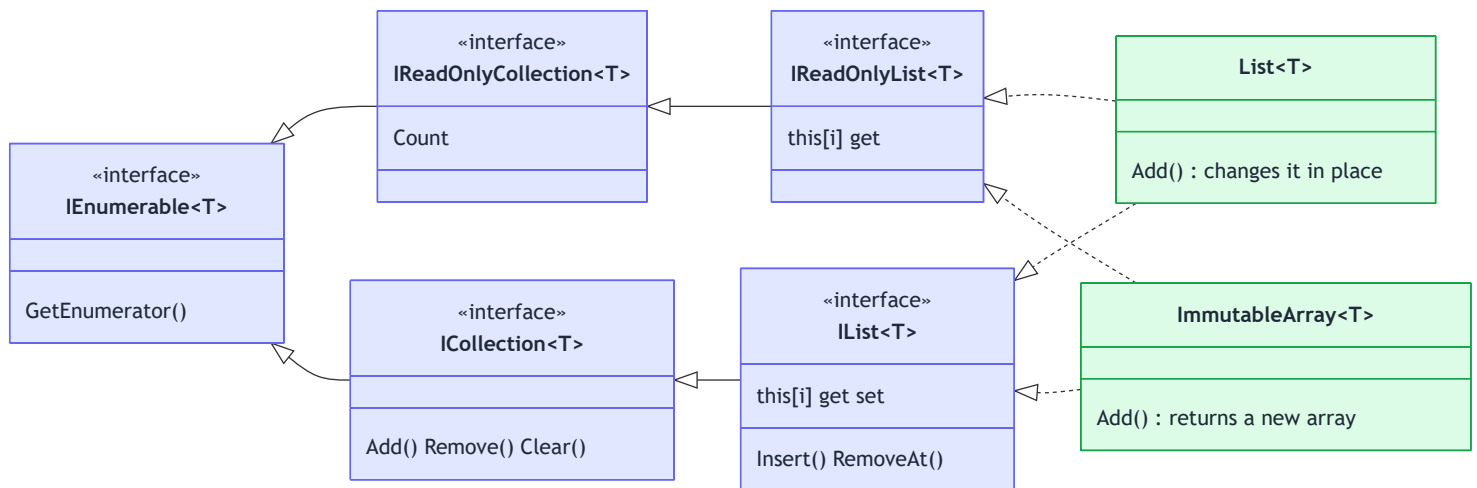
Pick a collection type by the promise it makes to the caller, not by the data it holds. `IEnumerable<T>` promises only that you can iterate — perhaps lazily, perhaps expensively. `IReadOnlyList<T>` promises a count and an indexer over data that already exists. `ImmutableArray<T>` and `FrozenDictionary<TKey, TValue>` promise that nothing will change. Kotlin's read-only `List<T>` is the `IReadOnlyList<T>` idea; Java has no read-only interface, only wrappers that throw.

Read-only is a view; immutable is a guarantee. In 2.3 one list is exposed four ways and then grows. The interface view and the `AsReadOnly()` wrapper see the new item; the immutable and frozen copies do not. A caller holding the `IReadOnlyList<T>` view can even cast it back to `List<T>` and remove an item — it compiles. The `AsReadOnly()` wrapper is a different type, so that cast fails; it still cannot stop the owner's changes.

The two interface families are separate. `IList<T>` extends `ICollection<T>` and `IEnumerable<T>` only — not `IReadOnlyList<T>` ([IList<T>](#)). Concrete types such as `List<T>`, arrays and `ImmutableArray<T>` implement both, so a method that asks for `IReadOnlyList<T>` accepts all of them.

Collection expressions are one literal syntax for all of them. C# 12 (November 2023, [The history of C#](#)) added `[a, b, .. rest]`, which converts to arrays, spans, `List<T>`, immutable types and the collection interfaces. For an interface target the compiler chooses the concrete type ([Collection expressions](#)): the sample's `IReadOnlyList<string>` is *not* a `List<string>`.

2.1 · Two interface families — what each type promises



2.2 · Collection expressions — one literal for every target

```
C# samples/L04.Collections/Program.cs · #region collection-expressions

int[] claimFreeYears = [0, 1, 2, 3, 4, 5];
List<string> makes = ["Toyota", "Honda"];
string[] shortlist =[.. makes, "Isuzu", .. makes.Take(1)];
IReadOnlyList<string> channels = ["SMS", "Email", "LINE"];

Console.WriteLine($" shortlist {string.Join(", ", shortlist)}");
Console.WriteLine($" years      {claimFreeYears.Length} elements");
Console.WriteLine($" channels is a List<string>? {channels is List<string>}");
Console.WriteLine($" channels type {channels.GetType().Name}");
```

The spread .. inlines any enumerable, including a LINQ query. The output is Toyota, Honda, Isuzu, Toyota and 6 elements. The last two lines print False and <z__ReadOnlyArray`1: for an IReadOnlyList<T> target this compiler synthesised its own read-only type, so a caller cannot cast it to List<T> and add. The type name is a compiler detail, not a contract. Unlike a LINQ query, a collection expression is always evaluated into memory immediately ([Collection expressions](#)).

2.3 · Read-only is not immutable

```
C# · four views Program.cs · #region readonly-views

var classes = new List<string> { "Class1", "Class3" };
IReadOnlyList<string> view = classes; // same object
var wrapper = classes.AsReadOnly(); // live wrapper
var snapshot = classes.ToArray(); // copy
var frozen = classes.ToFrozenSet(); // copy, fast reads

classes.Add("Class2Plus");
Console.WriteLine($" view      {view.Count}");
Console.WriteLine($" wrapper  {wrapper.Count}");
Console.WriteLine($" snapshot {snapshot.Length}");
Console.WriteLine($" frozen   {frozen.Count}");

((List<string>)view).Remove("Class1"); // compiles!
Console.WriteLine($" after cast + Remove {view.Count}");
```

```
Output · part 2 L04.Collections, captured

2. read-only is not immutable
view      3
wrapper    3
snapshot   2
frozen     2
after cast + Remove  2
```

Only the copies are safe from the owner. view is the same object as classes, and AsReadOnly() wraps it live, so both report 3 after the Add. ToImmutableArray() and ToFrozenSet() copied, so they report 2. The last line shows why "read-only" is a courtesy: the cast back to List<string> removed an item through the view. The same cast on wrapper would throw InvalidCastException, but the wrapper still shows every later change. Return IReadOnlyList<T> from APIs, but hand out an immutable copy when the caller must not see — or cause — changes.

1 List<T> — the default mutable list C# Java/Kotlin • Same idea

A growable array: $O(1)$ index, amortised $O(1)$ append.

JAVA `ArrayList<E>`

KOTLIN `MutableList / ArrayList`

TS `Array<T>`

WATCH Adding inside foreach throws `InvalidOperationException` (3.4) — Java's `ConcurrentModificationException`

2 Dictionary<TKey,TValue> and HashSet<T> C# • Trap

Hash map and hash set; iteration order is not part of the contract.

JAVA `HashMap · HashSet`

KOTLIN `mutableMapOf · mutableSetOf`

TRAP `dict[key]` on a missing key **throws** `KeyNotFoundException` (4.4); Kotlin `map[key]` and Java `get` return null — use `TryGetValue`

ORDERED `OrderedDictionary<TKey,TValue>` is generic since .NET 9 ([What's new in .NET 9 libraries](#))

3 Arrays, Span<T> and ReadOnlySpan<T> C# • Different

Fixed-length arrays, and a stack-only window over an array, a string or stack memory.

JAVA `T[]` · a `ByteBuffer` slice

SPAN A ref struct: it cannot be a class field, be boxed or be captured by a lambda ([ref struct types](#))

USE WHEN Parsing and hot loops; not in domain models or LINQ pipelines

C# 14 Arrays convert to spans implicitly — which changed LINQ overload binding (7.4)

2.4 · The collection family (continued)

4 Immutable collections C# • Different

`ImmutableArray<T>`, `ImmutableList<T>`, `ImmutableDictionary` — every change returns a new instance.

JAVA `List.copyOf` and Guava `ImmutableList` are unmodifiable: `add` throws, where `ImmutableList.Add` returns a new list

KOTLIN `kotlinx.collections.immutable` persistent collections: `add` returns a new one

USE WHEN A snapshot shared across requests or threads

TRAP A record compares an `ImmutableArray<T>` by array reference, not by elements (5)

SAMPLE `ToImmutableArray()` kept 2 items after the source list grew (2.3)

5 Frozen collections (.NET 8) C# architecture • Different

`FrozenDictionary` and `FrozenSet`: pay once at creation, read faster for the life of the process.

JAVA `Map.copyOf` · Guava `ImmutableMap` — immutable snapshots

USE WHEN Start-up lookup tables read on every request: base rates by class, partner endpoints, feature flags

AVOID Anything rebuilt per request — creation is the expensive part

SAMPLE `baseRates` in `L04.Collections` is a `FrozenDictionary<CoverageClass, decimal>` · [What's new in .NET 8](#)

6 Read-only interfaces and wrappers C# Java/Kotlin • Trap

`IReadOnlyList<T>`, `AsReadOnly()`, `ReadOnlySet<T>` (.NET 9) — views over data someone else can still change.

KOTLIN `List<T>` over a `MutableList` — the same view semantics

JAVA `Collections.unmodifiableList` — a view that throws on write

TRAP A cast of the `IReadOnlyList<T>` view back to `List<T>` compiles and mutates (2.3); the `AsReadOnly()` wrapper refuses that cast

TRAP `IList<T>` does not extend `IReadOnlyList<T>` ([IList<T>](#))

2.5 · Choosing the type at an API boundary

Position	Use	Why	Habit it replaces
Parameter you only iterate	<code>IEnumerable<T></code>	Accepts arrays, lists and queries; enumerate it once	<code>Iterable<T></code>
Parameter you count or index	<code>IReadOnlyList<T></code>	No defensive copy, no second enumeration	Kotlin <code>List<T></code>
Result of a query method	<code>IReadOnlyList<T></code> , materialised	The caller cannot re-run your query by accident (3.2)	<code>stream().toList()</code>
A lazy stream you produce	<code>IEnumerable<T></code> iterator; <code>IAsyncEnumerable<T></code> for I/O — lesson 05 (Async, Tasks & Concurrency)	Say it is lazy in the name or the doc comment	Kotlin <code>Sequence · Flow</code>
Snapshot shared across requests	<code>ImmutableArray<T></code> , <code>ImmutableDictionary</code>	Changes return a new instance	Guava immutable collections
Built once, read on every request	<code>FrozenDictionary</code> , <code>FrozenSet</code>	Read-optimised; creation costs more	a static final <code>Map</code>
Private working state	<code>List<T></code> , <code>Dictionary<TKey, TValue></code>	Fastest to fill; never exposed	<code>ArrayList</code> , <code>HashMap</code>
Hot loop over a buffer	<code>Span<T></code> , <code>ReadOnlySpan<T></code>	No allocation; stack-only	<code>ByteBuffer</code>

3 · Deferred execution — a query is a recipe, not a result

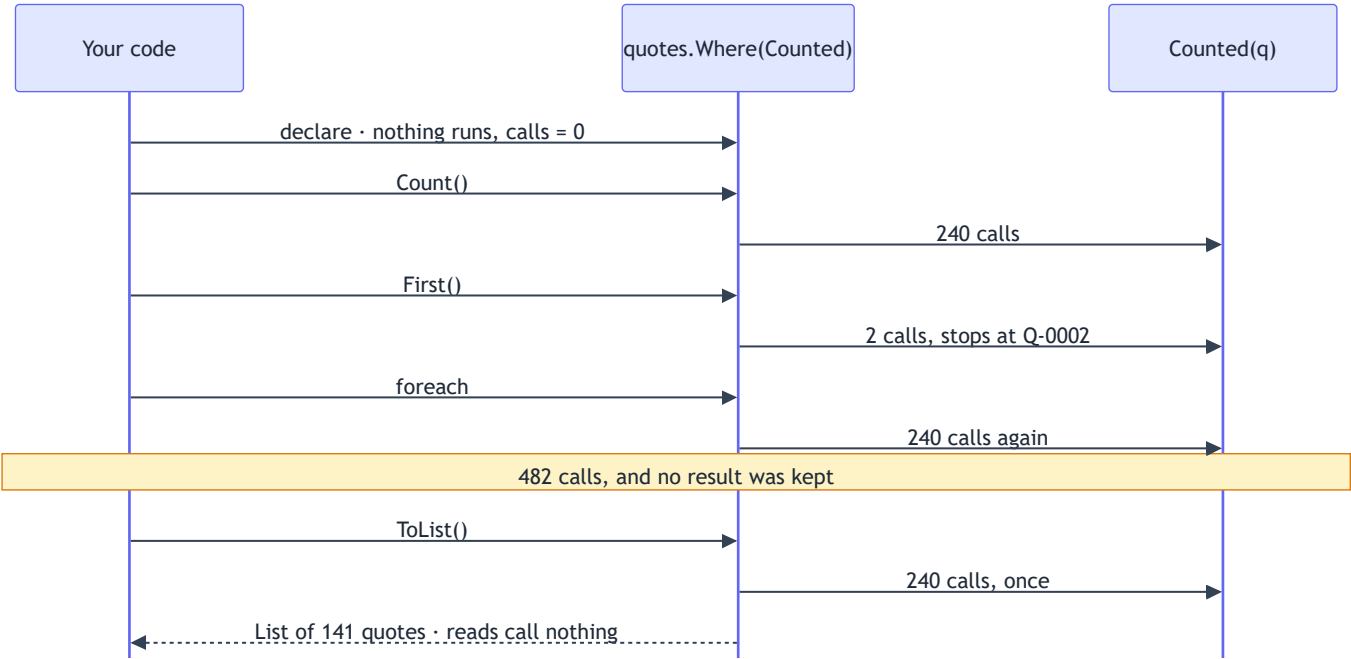
A LINQ query over `IEnumerable<T>` runs every time you enumerate it, and nothing stops you enumerating it twice. A Java Stream should be operated on only once and may throw `IllegalStateException` when reused ([java.util.stream.Stream](#)). Kotlin sequences can be iterated many times ([Kotlin sequences](#)), and LINQ follows Kotlin: reuse is legal, silent and re-executes every step. In the sample a `Count()`, a `First()` and a `foreach` over one query called its predicate **482** times for 240 quotes; `ToList()` once, then three reads, called it 240 times.

Materialise at the boundary. `ToList`, `ToArray`, `ToDictionary` and `ToLookup` execute immediately and keep the results; `Count`, `First` and `Sum` execute immediately and keep nothing; `Where` and `Select` are deferred and streaming; `GroupBy` and `OrderBy` are deferred but read the whole source each time they run ([Introduction to LINQ queries](#)). A repository or service method that returns a deferred query hands its cost — and, with EF Core, its open context — to every caller; lesson 09 (Data Access with EF Core) covers that case.

Deferred also means late. A query reads captured variables and the source collection when it is enumerated, not when it is written: raising a threshold after declaring the query changed its count from 48 to 18. Changing a `List<T>` while a `foreach` walks it throws `InvalidOperationException`, as Java throws `ConcurrentModificationException`.

`yield return` is how you write your own lazy sequence. The compiler rewrites the method into a state machine whose body does not start until the first `MoveNext` — the output prints *iterator created before (iterator body starts)*. Kotlin's sequence `{ yield(x) }` is the same idea with a different keyword.

3.1 · What runs when — one query, four calls



the predicate count after each step is the sample's measured output · ToList is the only call whose result is kept · colours only separate the actors (indigo) from the summary note (amber)

3.2 · Reusing a query — Java Streams vs LINQ

Java — for comparison the habit you bring

```
// a Stream is single-use: reuse throws
int[] calls = {0}; // lambdas capture effectively final
Stream<Quote> accepted = quotes.stream()
    .filter(q -> { calls[0]++; return q.isAccepted(); });

long count = accepted.count(); // runs the pipeline
accepted.findFirst();
// IllegalStateException: stream has already
// been operated upon or closed

// to read results more than once, collect first
List<Quote> list = quotes.stream()
    .filter(Quote::isAccepted)
    .toList();
```

C# Program.cs · #region deferred

```
var calls = 0;
bool Counted(Quote q) { calls++; return q.IsAccepted; }

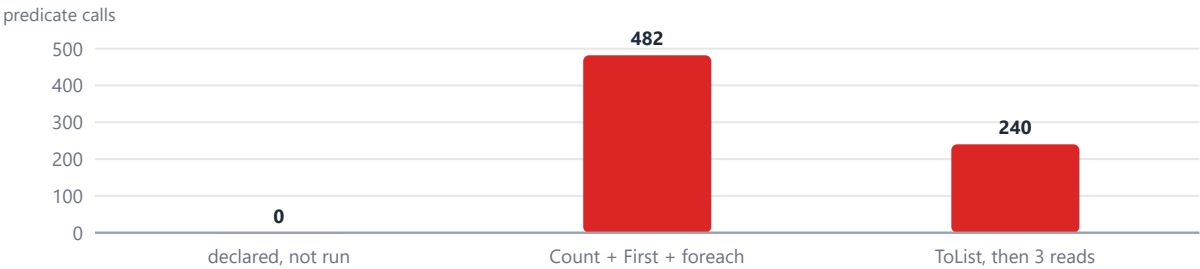
var accepted = quotes.Where(Counted); // method group -> Func
Console.WriteLine($" declared calls = {calls}");

var count = accepted.Count(); // pass 1: all quotes
var first = accepted.First(); // pass 2: stops early
foreach (var item in accepted) { } // pass 3: all again
Console.WriteLine($" three passes calls = {calls}");

calls = 0;
var list = accepted.ToList(); // one pass, then data
_ = (list.Count, list[0]);
foreach (var item in list) { }
Console.WriteLine($" ToList + 3 reads calls = {calls}");
```

Java fails loudly on reuse; C# re-runs quietly. Both pipelines are lazy, but a LINQ query is an `IEnumerable<T>` that starts over on every enumeration. The C# panel printed `calls = 0`, then `calls = 482`, then `calls = 240`. Against a database or a remote API, “quietly re-runs” means three round trips. Analyzer rule CA1851 flags possible multiple enumeration, but it is off by default (CA1851).

3.3 · Predicate calls for the same three reads



calls counted by the sample over the 240-quote book · measured from a real run

Materialising turns three passes into one. 482 = 240 for Count + 2 for First, which stops at Q-0002, + 240 for the foreach. Measured
from L04.Collections, and asserted by a test in L04.QuoteData.Tests.

3.4 · Deferred means late: captured variables and changing sources

C# · captured	Program.cs · #region captured	C# · changed list	Program.cs · #region modified
<pre>var threshold = 10_000m; var pricey = quotes.Where(q => q.Total.Amount > threshold); Console.WriteLine(\$" > threshold, run 1 {pricey.Count()}"); threshold = 12_000m; // the lambda captured the VARIABLE Console.WriteLine(\$" > threshold, run 2 {pricey.Count()}");</pre>		<pre>var queue = new List<string> { "Q-0001", "Q-0002" }; try { foreach (var id in queue) if (id == "Q-0001") queue.Add("Q-0003"); } catch (InvalidOperationException ex) { Console.WriteLine(\$" {ex.GetType().Name}"); }</pre>	

The lambda captured the variable threshold, not its value. The same query printed 48, then 18 after the assignment. On the right, Add during enumeration prints InvalidOperationException: List<T> enumerators check a version number, like Java’s fail-fast iterators. Iterate a copy (queue.ToList()) when the loop must change the source.

3.5 · Writing a lazy sequence — Kotlin sequence {} vs C# yield return

Kotlin — for comparison	sequence builder	C#	Program.cs · #region yield
<pre>fun renewals(inception: LocalDate) = sequence { println(" (sequence body starts)") var d = inception while (true) { yield(d) d = d.plusYears(1) } } val renewals = renewals(LocalDate.of(2026, 11, 1)) println(" sequence created") renewals.take(3).forEach { println(" renewal \$it") }</pre>		<pre>var renewals = Renewals(new DateOnly(2026, 11, 1)); Console.WriteLine(" iterator created"); foreach (var date in renewals.Take(3)) Console.WriteLine(\$" renewal {date:yyyy-MM-dd}"); static IEnumerable<DateOnly> Renewals(DateOnly inception) { Console.WriteLine(" (iterator body starts)"); for (var d = inception; ; d = d.AddYears(1)) yield return d; }</pre>	

Same model, different keyword. Both produce an infinite, lazy sequence of renewal dates, and Take(3) stops it after three. The C# run printed iterator created before (iterator body starts): calling Renewals only builds the state machine. A consequence to remember — argument checks at the top of an iterator method also run late, on the first MoveNext, not at the call.

4 · The operator map — Streams, Kotlin and TypeScript in LINQ

Most of LINQ is the pipeline you know under SQL-flavoured names. filter is Where, map is Select, flatMap is SelectMany, reduce/fold is Aggregate. The query syntax (from q in quotes where ... select ...) compiles to the same method calls, and it covers only a handful of operators — everything else must be written as a method call (Standard query operators). Most C# teams write method syntax and keep query syntax for joins and nested from clauses; 6.2 shows the same left join in both forms.

Most operators added since .NET 6 are ones Kotlin had first — and code you review may predate them. .NET 6 added Chunk, DistinctBy, MinBy and MaxBy (What's new in .NET 6). .NET 9 added CountBy and AggregateBy, which aggregate by key without allocating the groups GroupBy builds, and Index (What's new in .NET 9 libraries); like every LINQ operator that returns a sequence they are deferred. .NET 10 added LeftJoin and RightJoin — SQL's joins, not Kotlin's — replacing the GroupJoin + DefaultIfEmpty idiom; neither has query syntax (Join operations). It also added Shuffle and InfiniteSequence (Enumerable.Shuffle · Enumerable.InfiniteSequence) and ships System.Linq.AsyncEnumerable in the platform, replacing the community System.Linq.Async package (System.Linq.AsyncEnumerable in .NET 10) — lesson 05 (Async, Tasks & Concurrency).

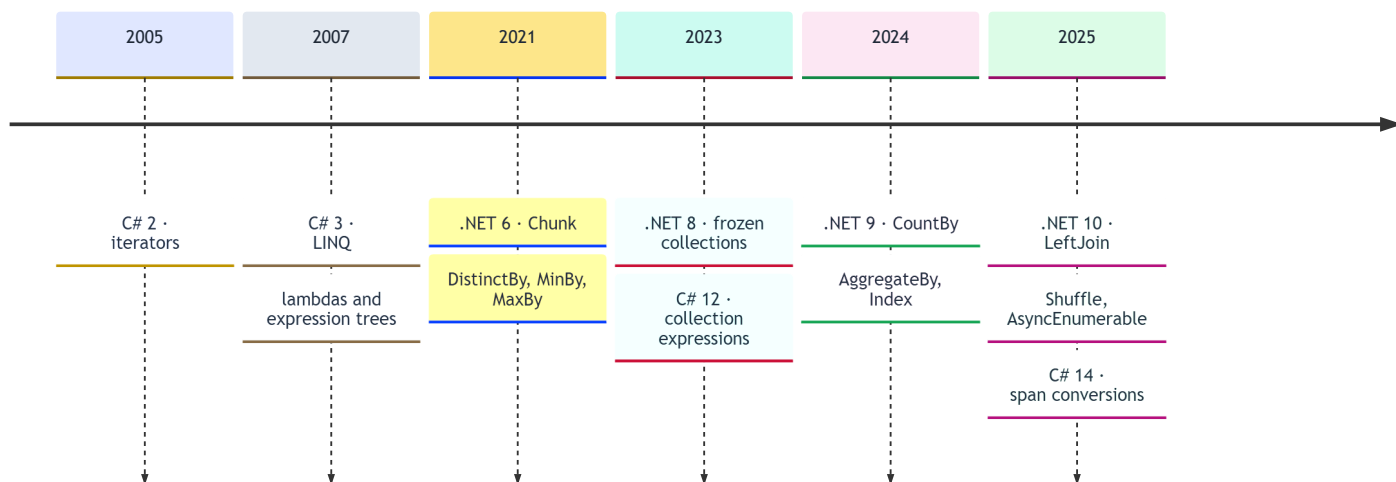
Know which operators execute now and which fail on a missing key. ToLookup runs immediately and GroupBy is deferred — the test GroupBy_is_deferred_and_ToLookup_is_immediate counts the key selector at 0, 240 and 720. A lookup returns an empty group for a key it has never seen; a dictionary indexer throws.

4.1 · Concept map — pipelines you know → LINQ

You already know	In .NET	Kind	Watch for
filter · arr.filter	Where	• Renamed	Deferred and streaming
map · arr.map	Select	• Renamed	Select((x, i) => ...) passes the index

You already know	In .NET	Kind	Watch for
flatMap	SelectMany	• Renamed	Also a second from in query syntax
reduce / fold	Aggregate	• Renamed	Without a seed it throws on an empty sequence
Collectors.groupingBy · Kotlin groupBy	GroupBy / ToLookup	• Trap	GroupBy is deferred and regroups on every enumeration; Kotlin's groupBy returns a map immediately — that is ToLookup
groupingBy(counting()) · Kotlin eachCount()	CountBy (.NET 9)	• Trap	Deferred: it recounts on every enumeration and yields pairs, not a map — add ToDictionary() where Kotlin gave you a Map; no row for an absent key
Kotlin groupingBy { }.fold	AggregateBy (.NET 9)	• Trap	Deferred like CountBy: one accumulator per key, no intermediate groups, rebuilt on every enumeration; Kotlin's fold returns a Map at once
Kotlin withIndex() · arr.entries()	Index() (.NET 9)	• Renamed	Yields (Index, Item) tuples
Kotlin chunked(n)	Chunk(n) (.NET 6)	• Renamed	Each chunk is an array; the last one holds the remainder
Kotlin distinctBy	DistinctBy (.NET 6)	• Same idea	Compares by a key without sorting; older code emulates it with GroupBy + First
Kotlin maxByOrNull	MaxBy (.NET 6)	• Trap	On an empty sequence MaxBy returns null only for reference types; for structs (including CountBy's KeyValuePair rows) it throws InvalidOperationException
Kotlin zip · Guava Streams.zip	Zip	• Renamed	Stops at the shorter sequence; the overload without a result selector yields tuples
Collectors.toMap · Kotlin associateBy	ToDictionary	• Trap	Duplicate keys throw ArgumentException, as toMap throws IllegalStateException; Kotlin's associateBy keeps the last one
map.get(key) returning null	dict[key] · lookup[key]	• Trap	The dictionary indexer throws KeyNotFoundException; a lookup returns an empty group
SQL LEFT JOIN	LeftJoin (.NET 10)	• Different	Before .NET 10: GroupJoin + SelectMany + DefaultIfEmpty
Stream single use	Re-enumerable IEnumerable<T>	• Trap	Every enumeration re-runs the whole pipeline (3.2)
toList() · spread into an array	ToList() · ToArray()	• Same idea	The point where the query actually runs
Kotlin sequence { yield() }	yield return	• Renamed	The method body starts on the first MoveNext
parallelStream()	PLINQ AsParallel()	• Different	lesson 05 (Async, Tasks & Concurrency)

4.2 · When each feature arrived — dating the LINQ in a codebase



language features and library operators by release year · C# 12 shipped with .NET 8 and C# 14 with .NET 10 · colours only separate the releases · verified against the C# history, the .NET what's-new pages and the API reference

4.3 · AggregateBy — a fold per key without building groups

C# samples/L04.QuoteData/QuoteDashboard.cs · #region net9-additions

```
// accepted premium per coverage class in one pass: no intermediate IGrouping objects
public static IEnumerable<KeyValuePair<CoverageClass, decimal>> PremiumByClass(
    IEnumerable<Quote> quotes) =>
    quotes
        .Where(q => q.IsAccepted)
        .AggregateBy(
            keySelector: q => q.Coverage,
            seed: 0m,
            (total, q) => total + q.Total.Amount)
        .OrderBy(kv => kv.Key);
```

This is Kotlin's `groupBy { }.fold`. One pass keeps one running total per coverage class; `GroupBy` would first collect every quote into a group and then sum each group. The dashboard prints `Class1 198,362 ... Class3 84,600 THB` (illustrative rates). Like `CountBy` it returns a deferred sequence of `KeyValuePairs` that is rebuilt on every enumeration (the test `CountBy_and_AggregateBy_are_deferred...` counts it), so sort explicitly when order matters and add `ToList()` or `ToDictionary()` to keep the result.

4.4 · A key that is not there — lookup vs dictionary

C# · missing key Program.cs · #region missing-key

```
var byMake = quotes.ToLookup(q => q.Vehicle.Make);
var soldByMake = quotes.CountBy(q => q.Vehicle.Make)
    .ToDictionary();

var teslas = byMake["Tesla"].Count();
Console.WriteLine($" lookup {teslas} Tesla quotes");
try { _ = soldByMake["Tesla"]; }
catch (KeyNotFoundException ex)
{
    Console.WriteLine($" dictionary {ex.GetType().Name}");
}
```

Output · part 9 L04.Collections, captured

```
9. a key that is not there
lookup      0 Tesla quotes
dictionary  KeyNotFoundException
```

Two grouped structures, two answers to a missing key. The `ILookup` from `ToLookup` returns an empty sequence for `Tesla`; the dictionary built from `CountBy` throws `KeyNotFoundException`. Kotlin and Java return null from `map[key]/get`, so ported code that null-checks the result crashes instead. Use `TryGetValue` or `GetValueOrDefault`.

5 · Functional C# — delegates, closures and immutable records

A C# lambda becomes a delegate, and delegate types are nominal, not structural. In Kotlin and TypeScript any function with the right shape fits a function type. In C#, `Func<Quote, bool>` and `Predicate<Quote>` have the same shape and still do not convert — the compiler reports CS0029, and the sample converts through `.Invoke` instead. Use `Func` and `Action`

everywhere, and declare a named delegate only where the name documents intent, as Adjustment does below ([Lambda expressions](#)).

Rules as values is the functional shape of a rating engine. Each rating rule is a named function in a list, and the net premium is a fold of the rules over the base premium: no mutable state, same input, same output. That is what makes the rules trivially testable — the young-driver theory asserts a pure function — and easy to reorder or extend. The rates are illustrative.

Closures capture variables, not values — and C#'s two loops differ. A for loop has one loop variable shared by every lambda created in it, so three lambdas print 3,3,3; a foreach gets a fresh variable per iteration and prints 0,1,2. TypeScript's for (let ...) creates a fresh binding per iteration, so the TypeScript habit is exactly wrong for a C# for loop. A static lambda cannot capture at all, which makes the intent explicit.

Immutability comes from records and with — shallow, like Kotlin's copy(). with copies the reference to a List<string>, so the renewal and the draft share one list. A record meant to be a value can hold an ImmutableArray<T> or another immutable collection, which stops the mutation — but not the surprise in equality: record equality compares an ImmutableArray<T> by its array reference, so two records with the same elements in separate arrays are *not* equal (the test A_record_holding_an_ImmutableArray... proves it). Override Equals with SequenceEqual when a record must compare by content. Records themselves are lesson 03 (Types, OOP & Generics).

5.1 · Rating rules as a list of functions, folded with Aggregate

C#

samples/L04.QuoteData/Rating.cs · #region rules-as-functions

```
// a rule is a value: takes the input and the running premium, returns a new premium
public delegate decimal Adjustment(RatingInput input, decimal premium);

public static readonly IReadOnlyList<(string Name, Adjustment Apply)> Rules =
[
    ("young driver", (r, p) => r.DriverAge < 25 ? p * 1.20m : p),
    ("claims loading", (r, p) => p * (1m + 0.15m * r.Claims)),
    // licence years stand in for claim-free years in this illustrative rule
    ("no-claim bonus", (r, p) => r.Claims == 0 ? p * NoClaimFactor(r.LicenceYears) : p),
    ("commercial use", (r, p) => r.Use == VehicleUse.Commercial ? p * 1.25m : p),
];

// fold the rules over the base premium: no mutable state, same input -> same output
public static decimal Net(RatingInput r) =>
    Rules.Aggregate(Base(r), (premium, rule) => rule.Apply(r, premium));

static decimal NoClaimFactor(int years) => 1m - 0.05m * Math.Min(years, 5);
```

Read Net as `rules.fold(base) { premium, rule -> rule(r, premium) }`. Adjustment is a named delegate type; each tuple pairs a rule name with a lambda. The collection expression builds an IReadOnlyList of tuples. Adding a commercial-fleet rule is one more line, and every rule can be tested alone: `Rating.Rules.Single(r => r.Name == "young driver").Apply`. Illustrative rates, not a real tariff.

5.2 · Closures in loops — TypeScript let vs C# for

TypeScript — for comparison

closures in loops

```
const byLet: Array<() => number> = [];
for (let i = 0; i < 3; i++)
    byLet.push(() => i);    // a fresh i per pass

const byVar: Array<() => number> = [];
for (var j = 0; j < 3; j++)
    byVar.push(() => j);    // ONE j, shared by all

const run = (fs: Array<() => number>) =>
    fs.map(f => f()).join(",");

console.log(run(byLet));    // 0,1,2
console.log(run(byVar));    // 3,3,3
```

C#

Program.cs · #region closures

```
var byFor = new List<Func<int>>();
for (var i = 0; i < 3; i++)
    byFor.Add(() => i);    // ONE i, shared by all

var byForeach = new List<Func<int>>();
foreach (var year in new[] { 0, 1, 2 })
    byForeach.Add(() => year);    // a fresh year per pass

Console.WriteLine($" for      {Run(byFor)}");
Console.WriteLine($" foreach {Run(byForeach)}");

static string Run(List<Func<int>> fs) =>
    string.Join(",", fs.Select(f => f()));
```

C#'s for behaves like TypeScript's var, not like let. The C# run printed for 3,3,3 and foreach 0,1,2. This bites in exactly the code you write in fan-out loops: building a list of tasks or callbacks with an index. Copy the loop variable into a local inside the body, or use foreach over a range.

5.3 · Delegates are nominal; with copies are shallow

C# Program.cs · #region delegate-types

```
Func<Quote, bool> isAccepted = q => q.IsAccepted;
// Predicate<Quote> p = isAccepted; // error CS0029
Predicate<Quote> asPredicate = isAccepted.Invoke;

var counted = quotes.Count(isAccepted);
var found = quotes.ToList().FindAll(asPredicate).Count;
Console.WriteLine($" Count {counted}, FindAll {found}");
```

C# Program.cs · #region with-shallow

```
var draft = new DraftQuote("Q-0001", Notes: ["created"]);
var renewal = draft with { QuoteId = "Q-0002" };
renewal.Notes.Add("renewal"); // the SAME List<string>

var notes = string.Join(" | ", draft.Notes);
var same = ReferenceEquals(draft.Notes, renewal.Notes);
Console.WriteLine($" draft notes {notes}");
Console.WriteLine($" same list? {same}");
```

Two small surprises for a Kotlin engineer. Left: the commented line does not compile (CS0029, checked on the build SDK); `isAccepted.Invoke` is a method group that converts to any compatible delegate type. Both counts print 141. Right: the run printed `draft notes created | renewal` and `same list? True` — changing the renewal's notes changed the draft's.

6 · Quote analytics — the dashboard habit in LINQ

A dashboard query is group, aggregate, order, project — whether it feeds a BI tool or a LINQ pipeline. You have written a few hundred of them in SQL for a BI dashboard. Written in LINQ they are type-checked, refactorable with the domain model and unit-testable against a fixed dataset. `QuoteBook.Generate()` produces the same 240 MotorQuote quotes on every machine — vehicles, drivers, coverage classes and statuses derived arithmetically from the index — so every number below is reproducible.

Prefer CountBy and AggregateBy when you need one number per key. `GroupBy` is right when you need the members of each group or several aggregates at once, as the conversion query does (code in 8.1). `CountBy` emits no row for a key that has no items, so a dashboard that must show zero rows walks a fixed list and looks each key up — `BandOrder` in 6.1.

Pin the numbers with a test. `Dashboard_numbers_match_the_captured_lesson_output` asserts the accepted counts, bands, top makes, premium totals and the 23 policies still pending issuance. A changed rating rule fails a test before a chart silently changes — the discipline of a query review, enforced by `dotnet test`.

6.1 · Premium bands — a switch expression inside CountBy

C# samples/L04.QuoteData/QuoteDashboard.cs · #region premium-bands

```
public static readonly string[] BandOrder = ["< 3k", "3k-6k", "6k-9k", "9k-12k", "12k+"];

public static string BandOf(Money premium) => premium.Amount switch
{
    < 3_000m => "< 3k",
    < 6_000m => "3k-6k",
    < 9_000m => "6k-9k",
    < 12_000m => "9k-12k",
    _ => "12k+",
};

// .NET 9 CountBy: one pass, one counter per key. A band with no quotes gets no key,
// so walk BandOrder and look each band up: an empty band shows 0, not a missing row
public static IReadOnlyList<KeyValuePair<string, int>> Bands(
    IEnumerable<Quote> quotes)
{
    var counts = quotes.CountBy(q => BandOf(q.Total)).ToDictionary();
    return BandOrder
        .Select(band => KeyValuePair.Create(band, counts.GetValueOrDefault(band)))
        .ToList();
}
```

The band is a pure function of the premium, so counting stays one line. `BandOf` uses relational patterns from lesson 02 (C# Language Essentials); `CountBy` counts per band in one pass, and `ToDictionary()` keeps the counts. A band with no quotes has no key, so `Bands` walks `BandOrder` and `GetValueOrDefault` supplies the 0 — the test `A_band_without_quotes_still_gets_a_row_with_zero` proves it. The method returns a materialised list, the rule from 2.5.

6.2 · Issuance — LeftJoin, and the query-syntax left join it replaces

C# · LeftJoin, .NET 10QuoteDashboard.cs · #region left-join

```
const string Unset = "(pending)"; // no policy issued yet

// .NET 10: accepted quotes, with a policy if one exists
public static IEnumerable<Issued> Issuance(
    IEnumerable<Quote> quotes,
    IEnumerable<Policy> policies) =>
    quotes
        .Where(q => q.IsAccepted)
        .LeftJoin(policies,
            q => q.QuoteId, p => p.QuoteId,
            (q, p) => (q.QuoteId, p?.PolicyNumber ?? Unset));
```

C# · query syntaxQuoteDashboard.cs · #region left-join-query

```
// before .NET 10: the same left join in query syntax
public static IEnumerable<Issued> IssuanceQuery(
    IEnumerable<Quote> quotes,
    IEnumerable<Policy> policies) =>
    from q in quotes
    where q.IsAccepted
    join p in policies on q.QuoteId equals p.QuoteId
        into issued
    from p in issued.DefaultIfEmpty()
    select (q.QuoteId, p?.PolicyNumber ?? Unset);
```

Both keep every accepted quote; a missing policy arrives as null. On the left, LeftJoin says it in one call. On the right, join ... into issued followed by from p in issued.DefaultIfEmpty() is the query-syntax shape of a left join — the GroupJoin + DefaultIfEmpty idiom you will meet in code written before .NET 10 (Join operations). The selector's second parameter is nullable either way, so p?.PolicyNumber ?? Unset is required, not defensive. The run reports 141 accepted and 23 pending, and the test The_C_sharp_query_syntax_left_join_matches_LeftJoin asserts the two return the same sequence.

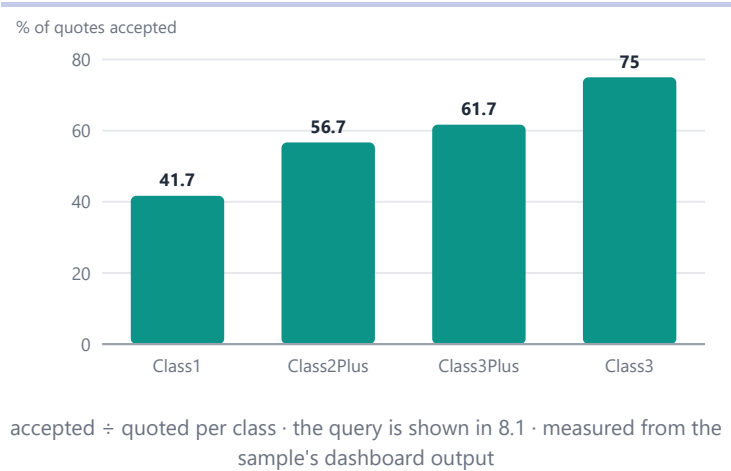
6.3 · What the dashboard prints

Output — L04.QuoteAnalytics, after the conversion and band lines charted in 6.4–6.5captured on the build machine

```
top makes, accepted quotes
1. Toyota    31
2. Honda     26
3. Isuzu     17
4. MG        17
5. Nissan    17
accepted premium by class, THB
Class1      198,362
Class2Plus  297,928
Class3Plus  348,568
Class3       84,600
issuance    141 accepted, 23 pending
export batches 100 + 100 + 40
lookup      49 Toyota quotes
```

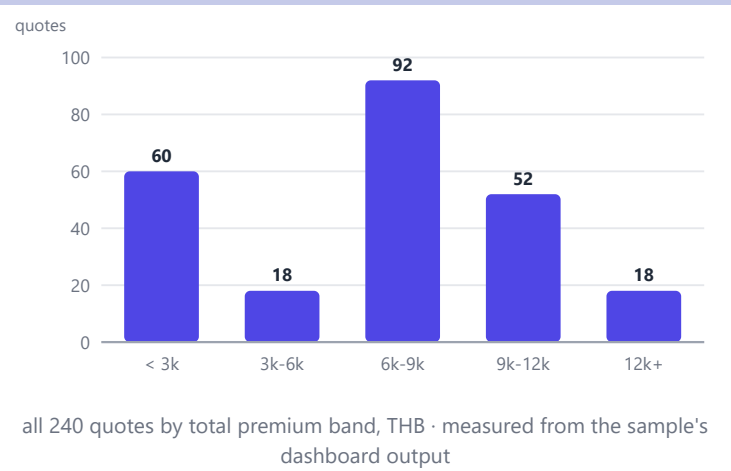
Every figure here except the batch sizes and the lookup count is asserted by a test. Rates are illustrative. Toyota leads accepted quotes; Isuzu, MG and Nissan tie at 17 and are ordered by name because the query adds ThenBy(kv => kv.Key) (code in 8.2). Chunk(100) split the book into export batches of 100 + 100 + 40.

6.4 · Conversion rate by coverage class



Acceptance rises as cover gets cheaper — 41.7% for Class 1 to 75.0% for Class 3. The dataset builds that gradient in on purpose, so the query has something to find: 141 of 240 quotes accepted overall. **Measured**, illustrative data.

6.5 · Quotes per premium band



The 6k–9k band holds 92 quotes; < 3k holds 60. Band edges are business choices, so the chart is only as meaningful as BandOf. The bars sum to 240 — the test Bands_account_for_every_quote checks that no quote falls outside a band. **Measured**, illustrative data.

7 · Expression trees — code as data, and a DynamoDB query generator

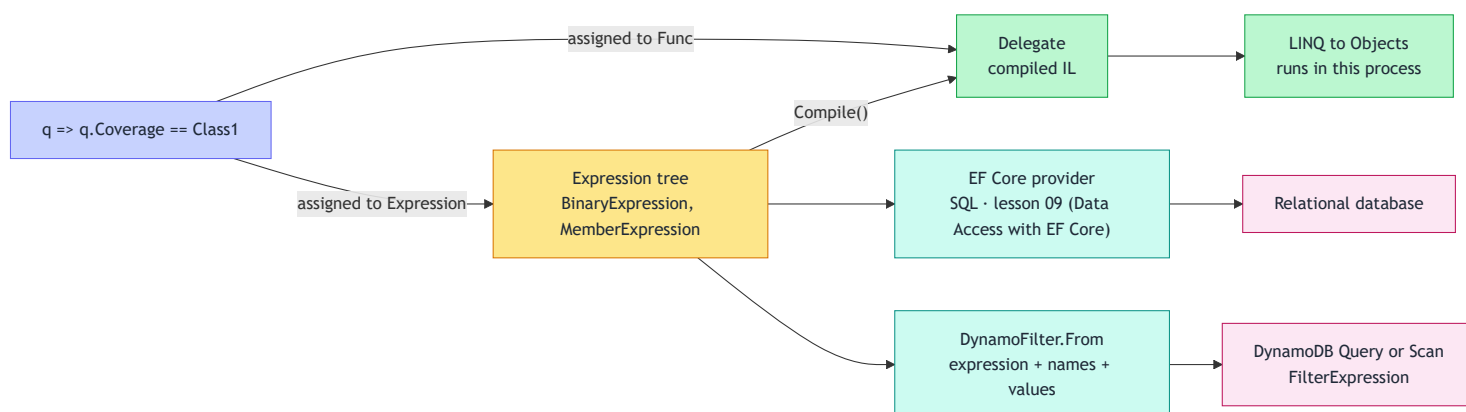
The same lambda compiles either to executable code or to a data structure, depending only on the type it is assigned to. Assigned to `Func<Quote, bool>` it becomes a delegate. Assigned to `Expression<Func<Quote, bool>>` the compiler emits code that builds a tree of nodes describing it ([Expression trees](#)). `IQueryable<T>` operators take expressions, which is how EF Core turns a C# predicate into SQL — lesson 09 (Data Access with EF Core).

A query generator is a translator over that tree. You built a DynamoDB query auto-generator before; in C# the compiler hands you the parsed predicate, and the library is one recursive switch over node shapes. `&&`, `||` and `!` become `AND`, `OR` and `NOT`; comparisons become comparators; `StartsWith` becomes `begins_with`; `Contains` becomes `contains` or `IN`. Every value goes through a `:value` placeholder, which DynamoDB always requires; every attribute goes through a `#name` placeholder, which DynamoDB requires only for reserved words and names with special characters — using it everywhere keeps the generator safe ([DynamoDB expression attribute names](#) · [DynamoDB condition and filter expressions](#)).

Compile time accepts what translation rejects. Any expression lambda that type-checks compiles, but a translator — yours or EF Core's — supports a subset. The computed property `IsAccepted` and a call to `ToUpperInvariant` both compile and both throw `NotSupportedException` at translation. The compiler itself refuses statement lambdas, `?.`, switch expressions, collection expressions, tuple literals and `is` patterns inside an expression tree; an interpolated string compiles, to a `string.Format` call that few providers can translate.

C# 14 changed what `array.Contains(x)` binds to inside an expression. With the new implicit span conversions it binds to `MemoryExtensions.Contains`, a `ReadOnlySpan<T>` method (7.4), and such trees throw when compiled with interpretation ([C# 14 overload resolution with span parameters](#)). The translator unwraps the conversion; for a provider you do not own, call `Enumerable.Contains` explicitly or cast to `IEnumerable<T>`.

7.1 · One lambda, two destinations



indigo = your source · green = code · amber = data the compiler builds · teal = a translator · rose = outside this process

7.2 · `Func<T,bool>` or `Expression<Func<T,bool>>`?

`Func<T, bool>` — a delegate

- Compiled IL; call it directly
- Used by `Enumerable` (LINQ to Objects)
- Any statement lambda, `?.`, switch expression or local function is allowed
- Opaque to a provider: it cannot be translated to SQL or a filter
- Choose it for in-memory rules — the rating rules in 5.1

`Expression<Func<T, bool>>` — a tree

- Data: nodes you can walk, rewrite or translate
- Used by `Queryable` — EF Core, your own generator
- Expression lambdas only; newer syntax is refused by the compiler
- Every provider supports a subset — unsupported calls fail at run time
- Choose it when the predicate must run somewhere else: SQL, DynamoDB, a search index

7.3 · Func vs Expression, a span surprise, and a generated filter

```
C# samples/L04.QuoteAnalytics/Program.cs · #region expression-demo

Func<Quote, bool> compiled = q => q.Coverage == CoverageClass.Class1;
Expression<Func<Quote, bool>> tree = q => q.Coverage == CoverageClass.Class1;
Console.WriteLine($"compiled {compiled.GetType().Name} -> {compiled(quotes[0])}");
Console.WriteLine($"tree {tree.Body.NodeType} -> {tree.Body}");

string[] shortlist = ["Toyota", "Honda"];
Expression<Func<Quote, bool>> inList = q => shortlist.Contains(q.Vehicle.Make);
var call = (MethodCallExpression)inList.Body;
Console.WriteLine($"binds to {call.Method.DeclaringType!.Name}.{call.Method.Name}");

var minPremium = 15_000m;
var filter = DynamoFilter.From<Quote>(q =>
    q.Coverage == CoverageClass.Class1
    && q.Total.Amount >= minPremium
    && q.Vehicle.Make.StartsWith("To"));
Console.WriteLine($"FilterExpression {filter.Expression}");
Console.WriteLine($"Names {string.Join(", ", filter.Names)}");
Console.WriteLine($"Values {string.Join(", ", filter.Values)}");
```

Three experiments in one region. The first pair declares the same lambda twice. The second shows what C# 14 binds `shortlist.Contains` to. The third hands a predicate with a captured `minPremium` to the generator. The output follows in 7.4.

7.4 · What the expression experiments print

Output — L04.QuoteAnalytics (expression part)

captured on the build machine

```
compiled Func`2 -> True
tree Equal -> (Convert(q.Coverage, Int32) == 0)
binds to MemoryExtensions.Contains
FilterExpression ((#n0 = :v0 AND #n1.#n2 >= :v1) AND begins_with(#n3.#n4, :v2))
Names [#n0, Coverage], [#n1, Total], [#n2, Amount], [#n3, Vehicle], [#n4, Make]
Values [:v0, Class1], [:v1, 15000], [:v2, To]
```

Read the second line slowly. The tree shows the enum comparison as `Convert(q.Coverage, Int32) == 0` — the compiler lowered it, which is why the translator strips `Convert` nodes and maps the number back to `Class1`. The captured `minPremium` arrives as the value 15000, evaluated when the filter was built. Five name placeholders cover one top-level attribute and two nested paths.

7.5 · The translator — pattern matching is the visitor

```
C# samples/L04.QueryGen/DynamoFilter.cs · #region translate

// one switch over node shapes: pattern matching is the visitor
public string Condition(Expression e) => e switch
{
    BinaryExpression { NodeType: ExpressionType.AndAlso } b
        => $"({Condition(b.Left)} AND {Condition(b.Right)})",
    BinaryExpression { NodeType: ExpressionType.OrElse } b
        => $"({Condition(b.Left)} OR {Condition(b.Right)})",
    UnaryExpression { NodeType: ExpressionType.Not } u
        => $"NOT ({Condition(u.Operand)})",
    BinaryExpression b when Comparators.TryGetValue(b.NodeType, out var op)
        => Compare(b.Left, op, b.Right),
    MethodCallExpression { Method.Name: "StartsWith", Object: { } s } m when IsOnItem(s)
        => $"begins_with({Path(s)}, {Operand(m.Arguments[0], typeof(string))})",
    MethodCallExpression { Method.Name: "Contains" } m
        => Contains(m),
    MemberExpression m when m.Type == typeof(bool) && IsOnItem(m)
        => $"{{Path(m)}} = {Constant(true)}",
    _ => throw Unsupported(e),
};
```

One switch expression replaces a visitor class hierarchy. Property patterns match node shapes (`BinaryExpression { NodeType: ExpressionType.AndAlso }`), and anything unmatched throws. `DynamoFilter.From<T>` calls `Condition(predicate.Body)` and returns the text with its name and value maps. Bare boolean members become `= :v`, because `DynamoDB` cannot use a Boolean attribute on its own as a condition ([DynamoDB condition and filter expressions](#)). A member counts as a stored attribute when it is a field or an auto-property, so a hand-written getter such as `IsAccepted` is refused as computed C#.

A filter expression is not a query plan. DynamoDB applies a filter after Query has read the items, so the read capacity consumed is the same with or without it, and the 1 MB page limit applies before filtering ([DynamoDB Query filter expressions](#)). Generate filters for the residual conditions; design partition and sort keys for the access pattern.

7.6 · What the generator translates — each row is a passing test

C# predicate	Filter expression	Result
q.QuoteId == "Q-0001"	#n0 = :v0	✓ translated
r.Year < 2020 ... >=, !=	< <= > >= = <>	✓ translated
2020 < r.Year	#n0 > :v0 — constant moved right	✓ translated
(a b) && !(c)	((... OR ...) AND NOT (...))	✓ translated
q.Vehicle.Make.StartsWith("To")	begins_with(#n0.#n1, :v0)	✓ translated
q.Vehicle.Model.Contains("Revo")	contains(#n0.#n1, :v0)	✓ translated
makes.Contains(q.Vehicle.Make)	#n0.#n1 IN (:v0, :v1)	✓ translated
makes.Contains(...) with 0 or 101 values	— (DynamoDB: 1 to 100 values)	✗ refused
q.Coverage == CoverageClass.Class3Plus	value Class3Plus, not 2	✓ translated
r.Start >= from (a DateOnly)	value 2026-11-01	✓ translated
s.Make == "Toyota" (a get-only auto-property)	#n0 = :v0	✓ translated
a captured local changed after From	keeps the value read at translation	by design
q.IsAccepted (computed property)	—	✗ refused
Make.ToUpperInvariant() == "TOYOTA"	—	✗ refused
q.Vehicle.Make.EndsWith("a")	— (DynamoDB has no ends_with)	✗ refused
r.Tags.Contains("fleet") (a list attribute)	— (DynamoDB contains supports lists; this translator does not map it)	✗ refused

q is a Quote; r is the test file's own Row(Id, Renewal, Year, Start, Tags) record and s its Stored class. Every row is a test in `DynamoFilterTests.cs`; ✗ **refused** means the translator throws `NotSupportedException`.

7.7 · Testing the generator, including what it refuses

C# samples/L04.QueryGen.Tests/DynamoFilterTests.cs · #region tests

```
[Fact]
public void Captured_variables_are_read_when_the_filter_is_built()
{
    var min = 10_000m;
    Expression<Func<Quote, bool>> predicate = q => q.Total.Amount >= min;

    var built = DynamoFilter.From(predicate);
    min = 99_999m; // too late for `built`: it was translated already

    Assert.Equal("#n0.#n1 >= :v0", built.Expression);
    Assert.Equal(10_000m, built.Values[":v0"]);
    Assert.Equal(99_999m, DynamoFilter.From(predicate).Values[":v0"]);
}

[Fact]
public void A_computed_property_compiles_but_cannot_be_translated()
{
    var ex = Assert.Throws<NotSupportedException>(
        () => DynamoFilter.From<Quote>(q => q.IsAccepted));

    Assert.Contains("IsAccepted", ex.Message);
}
```

Test the translator's contract, not DynamoDB. No AWS SDK and no table: the tests compare expression text and placeholder maps. The first test here pins a subtle rule — a captured variable is read when `From` runs, so a filter built earlier keeps the old value. The second proves a predicate can compile and still be untranslatable. The project holds 24 test cases, covering every row of 7.6. **Measured**.

8 · Visual Basic LINQ — the richer query syntax

Visual Basic spells more of LINQ as clauses than C# does. The C# language reference lists eight clause keywords — from, where, select, group, into, orderby, join, let — plus six contextual words that only work inside them (in, on, equals, by, ascending, descending) ([C# query keywords](#)). Visual Basic has fourteen query clauses; six of them have no C# query keyword: Aggregate, Distinct, Skip, Skip While, Take and Take While ([Introduction to LINQ in Visual Basic](#)). A VB estate's queries read like SQL, and many never call a LINQ method by name.

Underneath it is the same LINQ. The clauses compile to the same Enumerable calls, deferred and immediate execution work the same way, and the VB console prints the same conversion and top-makes figures as the C# dashboard. An Aggregate clause that *begins* a query runs immediately and returns one value, or one object with a property per aggregate function; inside a From query it adds per-row aggregates and the query stays deferred ([Aggregate clause](#)).

Porting VB queries to C# is mechanical but not keyword-for-keyword. Group By ... Into Quoted = Count() becomes GroupBy plus a Select (8.1); Aggregate ... Into, Distinct, Skip While and Take While become method calls (8.3, 8.4); anonymous types become records or tuples when they cross a method boundary. Plan the port of a whole VB estate with lesson 06 (VB.NET for Legacy Estates).

8.1 · Conversion by class — C# methods vs VB Group By ... Into

C# · methods

QuoteDashboard.cs · #region conversion

```
public static IReadOnlyList<ClassConversion> Conversion(
    IEnumerable<Quote> quotes) =>
    quotes
        .GroupBy(q => q.Coverage)
        .OrderBy(g => g.Key)
        .Select(g => new ClassConversion(
            g.Key,
            Quoted: g.Count(),
            Accepted: g.Count(q => q.IsAccepted)))
        .ToList();
```

VB · Group By

Program.vb · #region conversion

```
Dim conversion =
    From q In quotes
    Group By q.Coverage Into
        Quoted = Count(),
        Accepted = Count(q.IsAccepted)
    Order By Coverage
```

VB names the aggregates inside the grouping clause. Count(q.IsAccepted) is VB's conditional count; C# writes g.Count(q => q.IsAccepted) inside a Select. The VB query has no ToList, so it stays deferred until the For Each that prints it — the C# method materialises before returning.

8.2 · Top makes — CountBy and Index vs a five-line VB query

C# · CountBy

QuoteDashboard.cs · #region top-makes

```
public static IEnumerable<MakeRank> TopMakes(
    IEnumerable<Quote> quotes, int take = 5) =>
    quotes
        .Where(q => q.IsAccepted)
        .CountBy(q => q.Vehicle.Make)
        .OrderByDescending(kv => kv.Value)
        .ThenBy(kv => kv.Key)
        .Take(take)
        .Index()
        .Select(x => new MakeRank(
            Rank: x.Index + 1, x.Item.Key, x.Item.Value));
```

VB · Group By

Program.vb · #region top-makes

```
Dim topMakes = From q In quotes
    Where q.IsAccepted
    Group By q.Vehicle.Make Into Sold = Count()
    Order By Sold Descending, Make
    Take 5
```

The same ranking, two idioms. C# uses the .NET 9 CountBy and Index to count and number in the pipeline; VB groups, orders by the named aggregate and Take 5, and numbers the rows in the loop that prints them. Both print Toyota 31, Honda 26, then Isuzu, MG and Nissan at 17.

8.3 · Aggregate ... Into — a clause C# does not have

C# · portQuoteDashboard.cs · #region summary

```
// C# port of Aggregate: one pass, three reads
public static Summary Summarise(IEnumerable<Quote> quotes)
{
    var amounts = quotes
        .Where(q => q.IsAccepted)
        .Select(q => q.Total.Amount)
        .ToList();
    return (amounts.Count, amounts.Sum(), amounts.Max());
}
```

VB · AggregateProgram.vb · #region aggregate

```
Dim book = Aggregate q In quotes
    Where q.IsAccepted
    Into Policies = Count(),
        Premium = Sum(q.Total.Amount),
        Largest = Max(q.Total.Amount)
```

VB's Aggregate runs immediately and returns one object. It returned Policies, Premium and Largest properties: 141 quotes, 929,457 THB, largest 16,759. C# has no such clause: the port materialises the amounts once with ToList and reads Count, Sum and Max from the list.

8.4 · Distinct, Skip While and Take While — clauses C# spells as methods

C# · portQuoteDashboard.cs · #region window

```
// C# port of Distinct, Skip While and Take While
public static IEnumerable<string> Makes(
    IEnumerable<Quote> quotes) =>
    quotes.Select(q => q.Vehicle.Make).Distinct().Order();

public static IEnumerable<Priced> Window(
    IEnumerable<Quote> quotes, decimal lo, decimal hi) =>
    quotes
        .OrderBy(q => q.Total.Amount)
        .SkipWhile(q => q.Total.Amount < lo)
        .TakeWhile(q => q.Total.Amount < hi)
        .Select(q => (q.QuoteId, q.Total.Amount));
```

VB · Skip WhileProgram.vb · #region distinct-skip-while

```
Dim makes = From q In quotes
    Select q.Vehicle.Make
    Distinct
    Order By Make

Dim window = From q In quotes
    Order By q.Total.Amount
    Skip While q.Total.Amount < 8400D
    Take While q.Total.Amount < 8600D
    Select q.QuoteId, q.Total.Amount
```

Both cut the same window from the premium-ordered book. OrderBy is stable, so the two quotes tied at 8,508.30 keep their book order (Q-0080, then Q-0143) in C# and in VB. Distinct and Order give the eight makes. The test The_C_sharp_ports_reproduce_the_captured_Visual_Basic_output asserts both ports against the figures the VB console printed.

8.5 · What the VB console prints

Output — L04.VbLinq, after the conversion and top-makes linescaptured on the build machine

```
accepted      141 quotes, 929,457 THB, largest 16,759
makes         BYD,Ford,Honda,Isuzu,Mazda,MG,Nissan,Toyota
premiums from 8,400 to under 8,600 THB
Q-0014      8,459.96
Q-0080      8,508.30
Q-0143      8,508.30
Q-0106      8,594.24
```

The conversion and top-makes lines, not repeated here, match the C# dashboard figure for figure: same records, same dataset, a different query language on the same runtime.

8.6 · Which operators have query syntax

Operator	C#	Visual Basic	Syntax
Where · Select · OrderBy	where · select · orderby	Where · Select · Order By	✓ both
GroupBy · Join · Let	group ... by ... into · join ... on ... equals · let	Group By ... Into · Join ... On ... Equals · Let	✓ both
Count or Sum per group	g.Count() inside select	Into Quoted = Count()	VB clause
Distinct · Skip · Take	.Distinct() · .Skip(n) · .Take(n)	Distinct · Skip n · Take n	VB clause
SkipWhile · TakeWhile	.SkipWhile(...) · .TakeWhile(...)	Skip While · Take While	VB clause
Aggregate a whole source	quotes.Count(...), .Sum(...), .Max(...)	Aggregate q In quotes Into Count(), Sum(...)	VB clause
CountBy · AggregateBy · LeftJoin	.CountBy(q => ...)	.CountBy(Function(q) ...)	methods

9 · Hands-on — build, test and run the lesson-04 samples

7 projects, one command; the 41 tests are the safety net. 24 pin the generator's translation rules and refusals; 17 pin the dataset, the dashboard figures, the C# ports of the VB queries, the pure rating rule, the deferred-versus-immediate behaviour of 3.3 and section 4, and the collection and record claims made in sections 2, 4 and 5 — so a changed rule fails a test before a captured panel here goes stale.

Try three changes, then run the tests. In `GroupBy_is_deferred_and_ToLookup_is_immediate` swap `GroupBy` and `ToLookup` and read which assertion fails; in the `readonly-views` region (2.3) add a second `classes.Add("Class3Plus")` and predict which of the four counts change before you run `L04.Collections`; teach `DynamoFilter` to emit `BETWEEN` for `r.Year >= 2020 && r.Year <= 2024` — DynamoDB has it ([DynamoDB condition and filter expressions](#)) — and add the test.

9.1 · Commands

Terminal

```
# build, test and run every lesson-04 project
python 0-script/verify_samples.py --only 4

# or one at a time, from the repository root
dotnet run --project lesson-04-linq-collections-functional/samples/L04.Collections
dotnet run --project lesson-04-linq-collections-functional/samples/L04.QuoteAnalytics
dotnet run --project lesson-04-linq-collections-functional/samples/L04.VbLinq
dotnet test lesson-04-linq-collections-functional/samples/L04.QueryGen.Tests
dotnet test lesson-04-linq-collections-functional/samples/L04.QuoteData.Tests
```

`verify_samples.py` must end with **7/7 sample projects passed**. The consoles need no input, network or database and exit within seconds.

9.2 · What the tests report

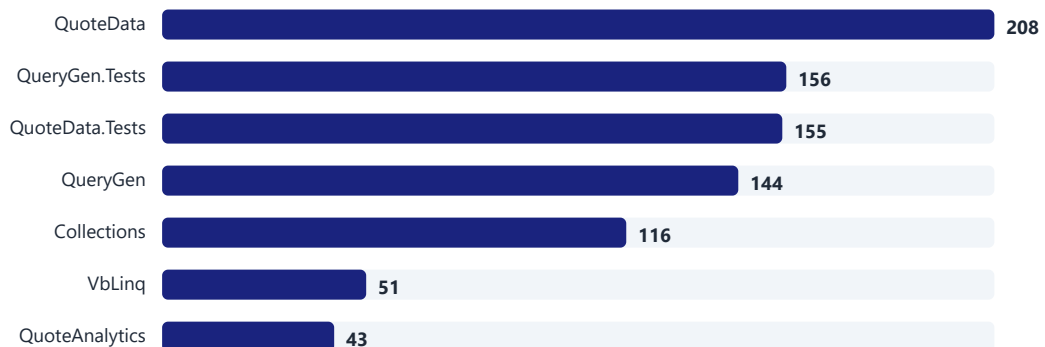
Output — dotnet test, durations omitted

captured on the build machine

```
Passed! - Failed: 0, Passed: 24, Skipped: 0, Total: 24 - L04.QueryGen.Tests.dll (net10.0)
Passed! - Failed: 0, Passed: 17, Skipped: 0, Total: 17 - L04.QuoteData.Tests.dll (net10.0)
```

41 tests pass: 24 in the generator project and 17 in the data project. • **Measured** from the two test files, and matching this run. If a rating rule or the dataset changes, `Dashboard_numbers_match_the_captured_lesson_output` fails first — re-capture the panels that quote those figures (3.3, 6.3 to 6.5).

9.3 · Where the lesson-04 code lives



non-blank, non-comment lines of C# and VB per sample project · measured when this PDF was built

The largest project is QuoteData (208 lines); the whole VB dashboard console is 51. The two test projects hold 311 lines and 41 test cases against 352 lines in the two libraries they test. • **Measured** with `loc()`; line count is a size signal, not a quality score.

⚠ Traps

1. **"A pipeline runs once."** A Java stream does; a LINQ query over `IEnumerable<T>` re-runs on every enumeration — 482 predicate calls for three reads in 3.3. Materialise with `ToList` before reading twice, and return materialised results from services.
2. **"Read-only means nobody can change it."** `ReadOnlyList<T>` and `AsReadOnly()` are views: the owner can still add, and a caller holding the `ReadOnlyList<T>` can cast it back to `List<T>` (2.3); the `AsReadOnly()` wrapper blocks that cast but not the owner's changes. Hand out `ImmutableArray<T>` or a frozen copy when that matters — and a record compares an `ImmutableArray<T>` by reference (5).
3. **"Loop variables are captured per iteration."** True for `foreach` and for TypeScript's `let`, false for a C# `for` loop: 3, 3, 3 in 5.2.
4. **"It compiled, so the provider can run it."** An expression tree type-checks C#, not the provider's subset: computed properties and unknown methods fail at translation (7.6), and in C# 14 `array.Contains` binds to a `span` method that interpreters reject.
5. **"map[key] returns null."** A `Dictionary` indexer throws `KeyNotFoundException` (4.4) and `ToDictionary` throws on duplicate keys (4.1).

✓ Checkpoint — you are ready for lesson 05 when

1. `python 0-script/verify_samples.py --only 4` reports 7/7 passed on your machine.
2. Given a method that returns `IEnumerable<T>`, you can say whether it is a query or data, and how many times a caller's code will run it.
3. You can choose the parameter and return type of a repository method from table 2.5 and say what each one promises its caller.
4. You can rewrite a Kotlin `groupBy + mapValues { it.value.size }` as `CountBy(...).ToDictionary()`, and a `GroupJoin/DefaultIfEmpty left join` as `LeftJoin`.
5. You can extend `DynamoFilter` with a new operator and a test, and say which predicates it must refuse.
6. You can read a VB `Group By ... Into` or `Aggregate ... Into` query aloud as C#.

🔍 Check yourself

1. A query quotes `.Where(p)` is counted, then its first item is read, then it is looped over. How many times does `p` run for 240 quotes whose second item is the first match, and which of `GroupBy` and `ToLookup` would have run immediately instead?
2. A service returns `classes.AsReadOnly()`. Can a caller see a later change, and can it cast the result back to `List<T>`? What do you return when the caller must not see changes?
3. Why does a `for` loop that adds `() => i` three times print 3, 3, 3, and what prints 0, 1, 2?
4. What is the difference between `Func<Quote, bool>` and `Expression<Func<Quote, bool>>` for the same lambda, and which one can EF Core translate?
5. Why does a `DynamoDB` filter expression not reduce the read capacity a `Query` consumes?
6. What does `shortlist.Contains(q.Vehicle.Make)` bind to inside an expression tree under C# 14, and how do you keep a provider working?
7. When does a VB `Aggregate` clause run immediately, and what C# does `Skip While` become?

Lesson 04 in one line: choose collections by the promise they make — read-only is a view, immutable is a guarantee; a LINQ query is a re-runnable recipe, so materialise at boundaries; delegates are nominal and `for` loops share their variable; the same lambda is code for `Enumerable` and data for `Queryable`, which is how you build a query generator; and Visual Basic says more of it in clauses.

Next: lesson 05 (Async, Tasks & Concurrency) — `Task`, `async/await`, cancellation and channels for fan-out to rating partners.